

Table S1 – Complete configuration of the preprocessing pipeline and data augmentation techniques applied using the Roboflow platform.

Parameter	Value
Final image resolution	640 × 640 pixels
Auto-orientation	Enabled
Resizing	Stretch
Horizontal Flip	50%
Rotation	±15°
Horizontal Shear	±15°
Vertical Shear	±15°
Scaling (Zoom)	±50%
Brightness	±25%
Saturation	−30% to +30%
Exposure	−25% to +25%
Noise	Not applied
Blur	Not applied
Crop	Not applied
Mosaic augmentation	Enabled
Random Erasing	Not applied
Applied to	Training set only

Note: All transformations were automatically applied by the Roboflow platform during the generation of the training dataset, while the validation and test datasets remained unchanged.

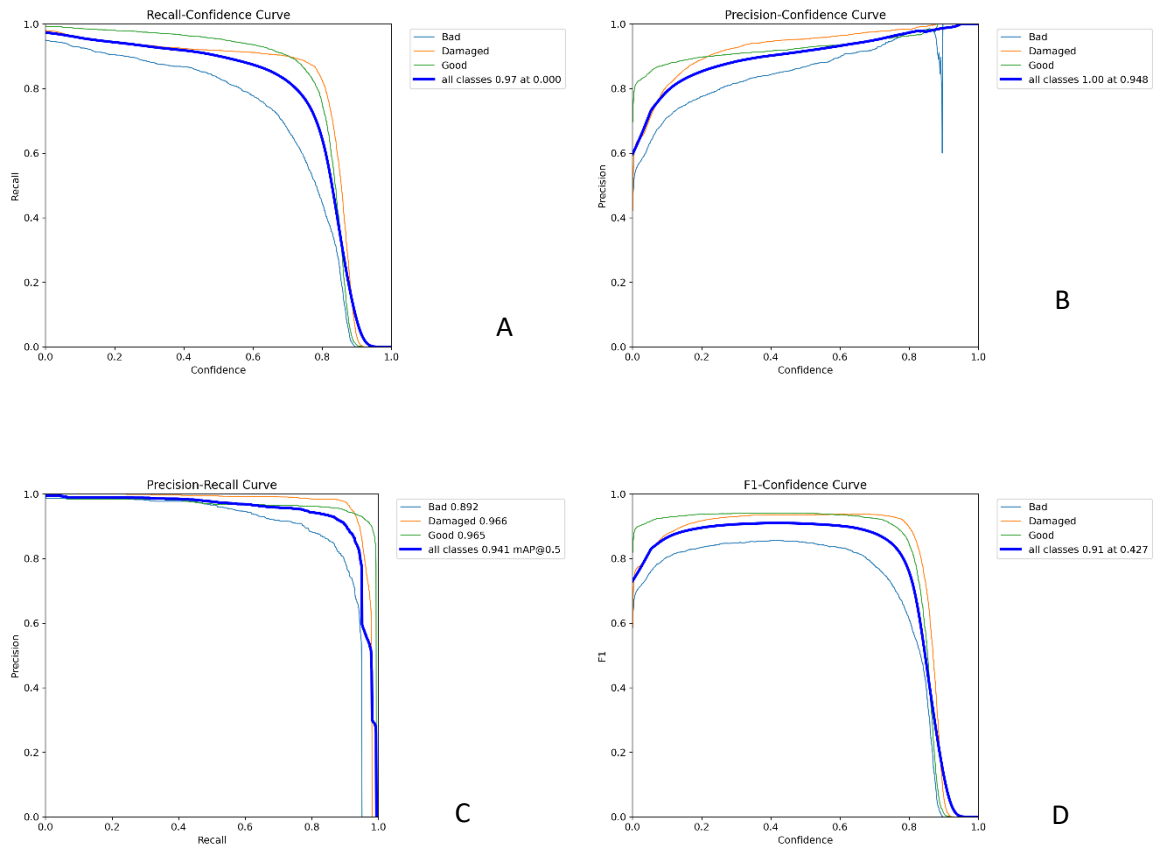


Figure S1 – Performance curves of the YOLOv8s model on the test dataset.

Legend: **(A)** Recall–Confidence curve; **(B)** Precision–Confidence curve; **(C)** Precision–Recall (PR) curve; and **(D)** F1–Confidence curve. These supplementary curves illustrate the model performance across different confidence thresholds and complement the quantitative evaluation presented in Table 4.

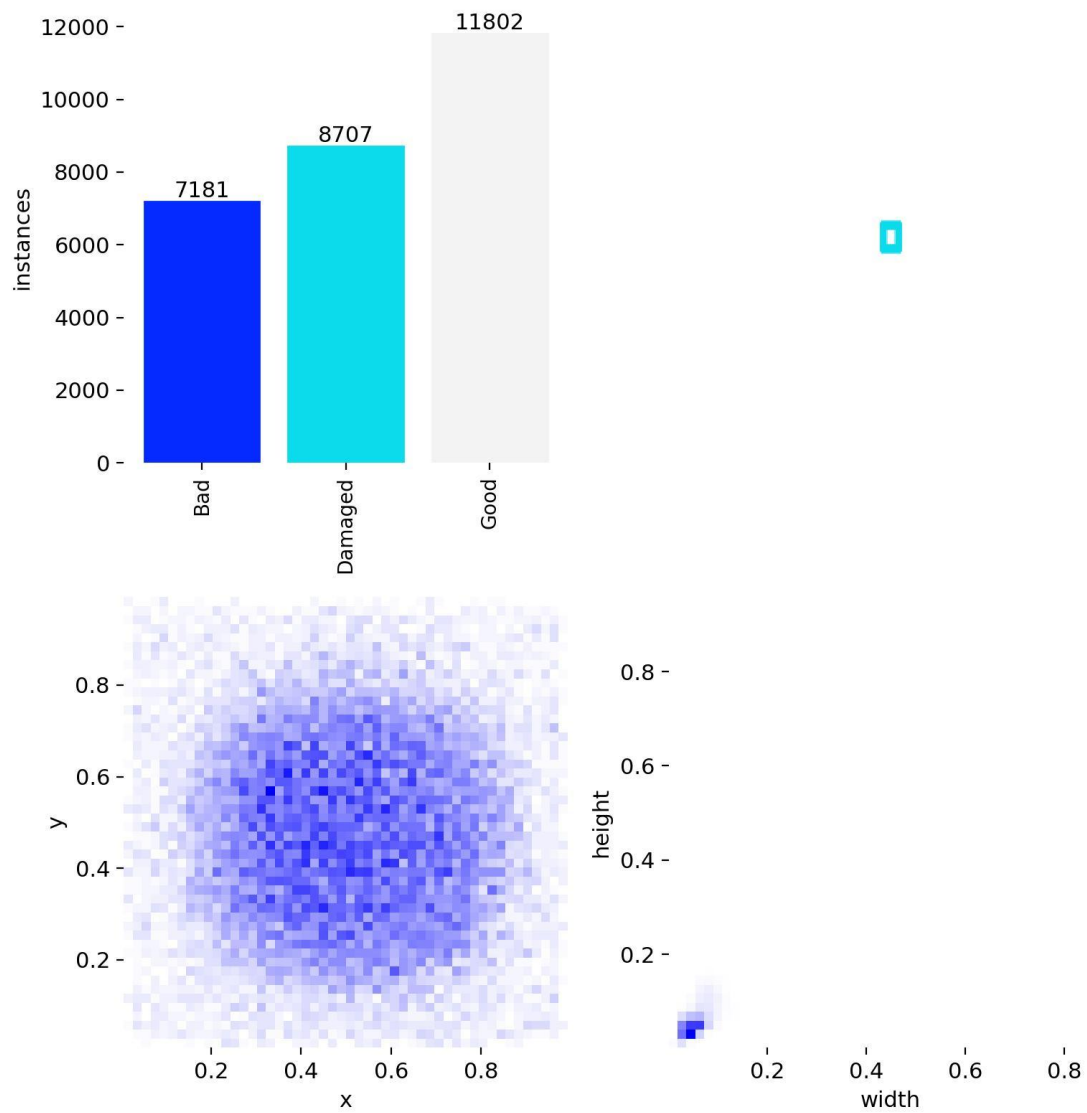


Figure S2 – Characterization of the dataset used for model training, showing the distribution of annotated instances across the three quality classes (Good, Damaged, and Bad), the spatial distribution of annotation centers (x, y), and the distribution of bounding box dimensions (width × height).

Table S2 – Detailed configuration of the computational environment, model architecture, and training hyperparameters used for the YOLOv8s model.

Category	Parameter	Configuration
Computational Environment	Platform	Google Colaboratory (Google Colab)
	Operating System	Linux (Google Colab environment)
	Programming Language	Python 3.12.13
	AI Framework	Ultralytics 8.4.75
	Deep Learning Library	PyTorch 2.11.0+cu128
	CUDA	CUDA 12.8
	OpenCV	Used for image processing
	NumPy	Used for numerical computations
	Pandas	Used for data manipulation
	Matplotlib	Used for data visualization and figure generation
Hardware	GPU	NVIDIA Tesla T4
	GPU Memory (VRAM)	15 GB (15,360 MiB)
	CPU	2 vCPUs (Google Colab)
	RAM	12.7 GB
	Available Storage	Approximately 46.9 GB
Model Architecture	Model	YOLOv8s
	Initial Weights	COCO (Transfer Learning)
	Input Image Resolution (imgsz)	640 × 640 pixels
	Batch Size	16
	Maximum Number of Epochs	300
	Early Stopping (patience)	100 epochs
	Optimizer	AdamW (Ultralytics default configuration) ¹
	Initial Learning Rate	Automatic (Ultralytics) ¹
	Weight Decay	Ultralytics default configuration ¹
	Learning Rate Scheduler	Automatic Ultralytics configuration ¹
	Random Seed	Ultralytics default (not manually fixed) ²
	Processing Device	CUDA:0 (NVIDIA Tesla T4 GPU)
Dataset	Dataset	Roboflow Universe – Corn Dataset
	Total Number of Images	621
	Total Number of Annotated Instances	34,124
	Number of Classes	3 (Good, Damaged, and Bad)
	Data Split	Training: 454 images (73%); Validation: 111 images (18%); Test: 56 images (9%)
Computational Performance	Total Training Time	Approximately 3.9 hours (224 completed epochs)
	Average Inference Time	11.4 ms per image (640 × 640 pixels)
	Computational Complexity	28.4 GFLOPs
	Number of Parameters	11,126,745
	Layers (fused model)	73
	Gradients	0 (inference mode)

Note: ¹Training was conducted using the default hyperparameter configuration of the Ultralytics YOLOv8 framework, without manual adjustment of the optimizer, learning rate, weight decay, or learning rate scheduler.

²No specific random seed was defined for model initialization; therefore, the default seed configuration provided by the Ultralytics framework was used.



Figure S3 – Ground-truth annotations for the second validation batch (val_batch2_label). Bounding boxes and class labels (Good, Damaged, and Bad) used as reference for evaluating the YOLOv8 model.

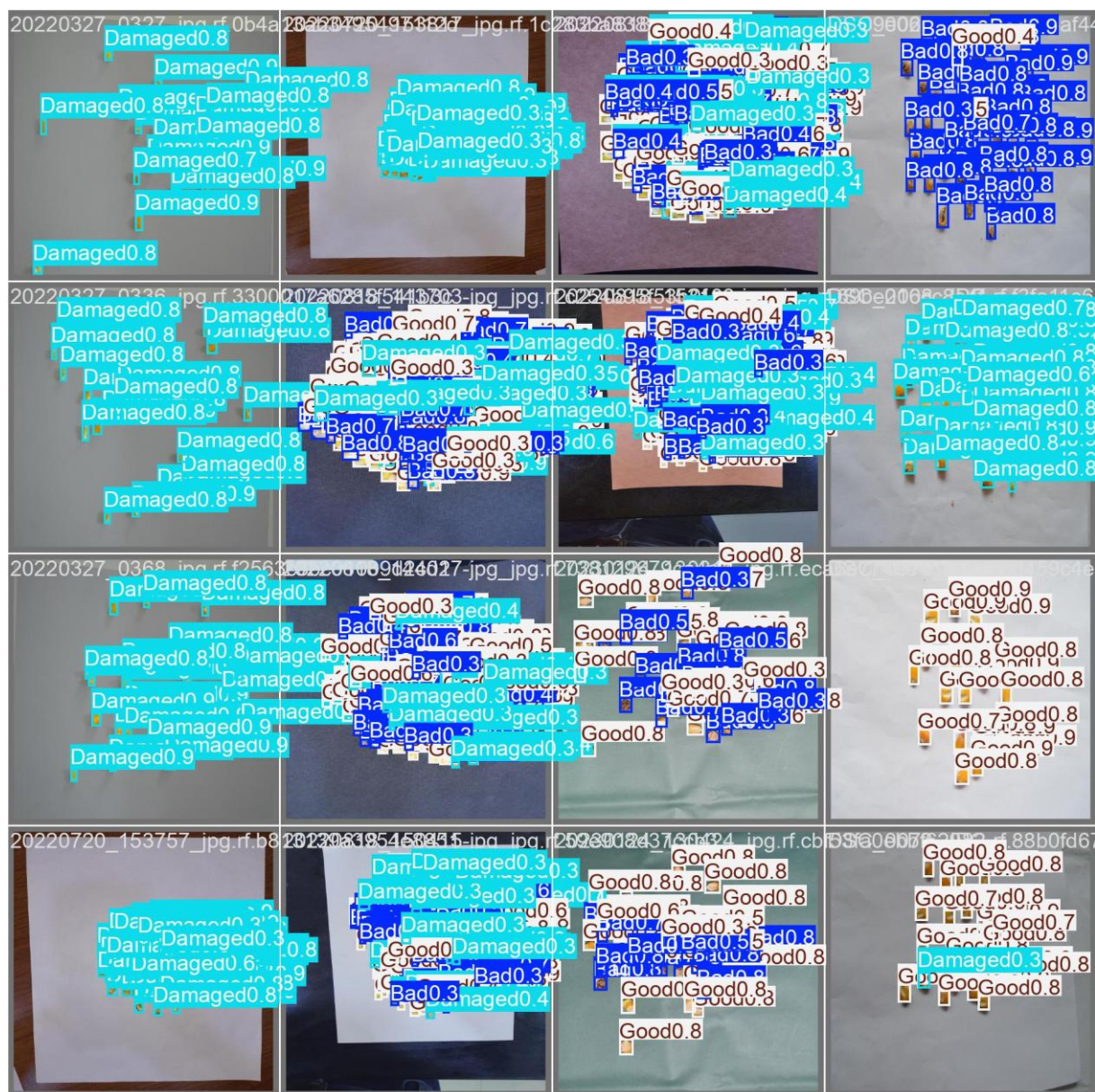


Figure S4 – Representative detection results obtained by the YOLOv8 model on the first validation batch (val_batch1_pred). Bounding boxes, predicted classes (Good, Damaged, and Bad), and confidence scores are displayed for each detected maize kernel